Rational Matrix Policy Optimization for Rational Feed-Forward Language Models

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Abstract

1 Introduction

2 Background and Related Work

3 Rational Matrix Policy Optimization

Let $\mathcal{L}(\theta)$ be the language-model training objective. For layer l, the RLB block projects the residual-stream input $x_l \in \mathbb{R}^d$ into G groups of width m,

$$z_l = A_l x_l, \qquad z_{l,g} = \mathrm{group}_g(z_l), \tag{1}$$

normalizes each group by its RMS radius,

$$r_{l,g} = \sqrt{m^{-1}\|z_{l,g}\|_2^2 + \epsilon_{\mathrm{rms}}}, \qquad u_{l,g} = z_{l,g}/r_{l,g}, \tag{2}$$

and applies a learned rational map in normalized coordinates,

$$h_{l,g} = r_{l,g} R_{l,g}(u_{l,g}), \qquad y_l = B_l \,\mathrm{concat}_{g=1}^{G} h_{l,g}. \tag{3}$$

The input matrix A_l selects rational domains, the coefficients of $R_{l,g}$ shape the group response, and the output matrix B_l recombines rational features. MatrixPolicy uses this separation by partitioning parameters as

$$\theta = \theta_{\mathrm{base}} \cup \theta_{\mathrm{coeff}} \cup \{A_l, B_l\}_{l=1}^{L}. \tag{4}$$

The base parameters use the same optimizer as the non-rational baseline. Rational coefficients use a coefficient rule. MatrixPolicy is applied only to A_l and B_l.

In the homogeneous case $\epsilon_{\mathrm{rms}} = 0$, and for nonzero group radii, each group has a positive scale gauge:

$$A_{l,g} \mapsto a_g A_{l,g}, \qquad B_{l,g} \mapsto a_g^{-1} B_{l,g}, \qquad a_g > 0. \tag{5}$$

The represented function is unchanged because $z_{l,g}$ and $r_{l,g}$ scale together, $u_{l,g}$ is unchanged, and the inverse scale in B_l cancels the scale in $h_{l,g}$. With a nonzero RMS floor, Appendix B bounds the resulting local gauge error under nondegenerate activations and treats the remaining implementation deviations through a residual model. This matters because optimizer state is stored in coordinates, not in the quotient space of represented functions.

At step t, each matrix role $r \in \{\mathrm{in}, \mathrm{out}\}$ receives a role-depth factor $\rho_r(l)$ and an on-policy statistic factor $s_{\mathrm{stat}}(l, r, t)$. The matrix update is

$$\begin{aligned} M_{l,r}^{t+1} = M_{l,r}^{t} &+ \Delta_{\mathrm{adapt}}(M_{l,r}; \eta_t a_{\mathrm{mat}}(l,r,t)[1 - \mu(l,r,t)]) \\ &+ \Delta_{\mathrm{mat}}(M_{l,r}; \eta_t a_{\mathrm{matnorm}} \mu(l,r,t)), \end{aligned} \tag{6}$$

where $M_{l,\mathrm{in}} = A_l$, $M_{l,\mathrm{out}} = B_l$, η_t is the shared base learning rate, Δ_{adapt} is an Adam-style adaptive step, and Δ_{mat} is a short early matrix-normalized step. The mixture coefficient

$\mu(l, r, t)$ is active only early in training. After the matrix update, MatrixPolicy computes a target log-ratio for each group and applies the bounded rebalance

$$A_{l,g} \leftarrow e^{\ell_g} A_{l,g}, \qquad B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}. \tag{7}$$

The update changes the representative used by the optimizer while preserving, or with $\epsilon_{\text{rms}} > 0$ approximately preserving, the represented block function. Full hyperparameter definitions are in Appendix A.

Algorithm 1: Rational Matrix Policy step. Given gradients $\nabla\mathcal{L}(\theta^t)$ and base learning rate η_t:

1. Update exponential moving averages of rational-group pressures.
2. Compute role-depth and group-stat multipliers for each RLB matrix.
3. Update non-rational parameters with the base optimizer and update rational coefficients with the coefficient rule.
4. Update each A_l and B_l with the MatrixPolicy mixture step.
5. Apply the bounded positive-gauge rebalance to each rational group.

4 Experiments

5 Discussion

6 Conclusion

A MatrixPolicy Details

Role-depth factors. For layers indexed by $l = 1, \ldots, L$ with $L > 1$, define normalized layer depth $d_l = (l-1)/(L-1)$ and set

$$\rho_{\text{in}}(l) = \text{clip}(1 - 0.50(d_l - 0.5), 0.55, 1.40), \tag{8}$$

$$\rho_{\text{out}}(l) = \text{clip}(1 + 1.00(d_l - 0.5), 0.55, 1.40), \tag{9}$$

$$a_{\text{role}}(l, r) = \max(0.10, 1 + 1.20(\rho_r(l) - 1)). \tag{10}$$

Let $\xi_t = t/T$ be training progress and let $\sigma_{a,b}(\xi)$ denote the smoothstep gate from 0 to 1 over $[a, b]$. The statistic multiplier $a_{\text{stat}}(l, r, t) > 0$ is the layer/role aggregate of the group preconditioners defined below, normalized to one when group statistics are not active. The local adaptive scale is

$$a_{\text{mat}}(l, r, t) = \text{clip}(3.0\, a_{\text{role}}(l, r) a_{\text{stat}}(l, r, t), 0.40, 4.0). \tag{11}$$

The matrix-normalized branch uses multiplier $a_{\text{matnorm}} > 0$ and mixture weight $\mu(l, r, t) \in [0, 1]$, a smoothstep-gated function of ξ_t, layer depth, matrix role, and $a_{\text{stat}}(l, r, t)$.

On-policy group statistics. All ratios and logarithms below are evaluated on strictly positive stabilized RMS, EMA, and gain quantities; the displayed formulas omit the corresponding epsilon floors. For group g, MatrixPolicy tracks relative gradient pressures

$$p_{\text{in},g} = \frac{\text{rms}(\nabla A_{l,g})}{\text{rms}(A_{l,g})}, \qquad p_{\text{out},g} = \frac{\text{rms}(\nabla B_{l,g})}{\text{rms}(B_{l,g})}, \qquad p_{\text{rat},g} = \text{rms}(\nabla\theta_{\text{coeff},g}). \tag{12}$$

With EMAs $q_{\text{in},g}$, $q_{\text{out},g}$, and $q_{\text{rat},g}$, we define

$$\pi_g = \log q_{\text{in},g} - \log q_{\text{out},g}, \qquad \alpha_g = \log q_{\text{rat},g} - \frac{1}{2}(\log q_{\text{in},g} + \log q_{\text{out},g}). \tag{13}$$

Group-stat preconditioning. The group preconditioner is centered and clipped:

$$c_g = c_{\mathrm{gain},g} c_{\mathrm{pressure},g} c_{\mathrm{activity},g}, \qquad c_g \leftarrow c_g / \operatorname{geomean}(c), \qquad c_g \leftarrow \operatorname{clip}(c_g, 0.75, 1.35). \quad (14)$$

Here $k_g > 0$ is the role-specific group gain used for matrix preconditioning: the derivative-RMS gain for A_l and the output-RMS gain for B_l. The factors are

$$c_{\mathrm{gain},g} = \left(\frac{\operatorname{geomean}(k)}{k_g}\right)^{0.20}, \qquad c_{\mathrm{pressure},g} = \exp(0.10\, d_{\mathrm{pressure},g}), \quad (15)$$

where $d_{\mathrm{pressure},g} = -\pi_g$ for A_l and $+\pi_g$ for B_l. The activity damping term is

$$c_{\mathrm{activity},g} = \exp[-0.20 \ \mathrm{ReLU}((\alpha_g - 0.05)/0.45)]. \quad (16)$$

Gauge rebalance. Let

$$\gamma_g = \log \operatorname{rms}(A_{l,g}) - \log \operatorname{rms}(B_{l,g}). \quad (17)$$

The target log-ratio is

$$\tau_g = \frac{1}{2}(\log d_g^{\mathrm{gain}} - \log o_g^{\mathrm{gain}}) + \beta_{\mathrm{pressure}}(\log q_{\mathrm{in},g} - \log q_{\mathrm{out},g}). \quad (18)$$

Here $d_g^{\mathrm{gain}} > 0$ and $o_g^{\mathrm{gain}} > 0$ are the derivative and output group-gain estimates, and $\beta_{\mathrm{pressure}}$ is the pressure balance coefficient. Let $s(t,l) \in [0,1]$ be the rebalance schedule, $a_g^{\mathrm{activity}} \in (0,1]$ the activity damping multiplier, and $\ell_{\max} > 0$ the maximum absolute log-scale move. The exact target step is $\ell_g^\star = \frac{1}{2}(\tau_g - \gamma_g)$, while the applied bounded move is

$$\ell_g = \operatorname{clip}(s(t,l) a_g^{\mathrm{activity}} \ell_g^\star, -\ell_{\max}, \ell_{\max}). \quad (19)$$

For true RMS values, or when stabilization floors are inactive, the clipped update maps γ_g to $\gamma_g + 2\ell_g$; with active RMS floors this identity is approximate and the discrepancy is included in the residual perturbation model. The update is therefore a bounded move toward τ_g rather than exact canonicalization unless clipping and schedule damping are inactive. The transform $A_{l,g} \leftarrow e^{\ell_g} A_{l,g}$ and $B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}$ preserves the homogeneous RLB function exactly under the nonzero-radius condition; the stabilized implementation is covered by the residual perturbation model in Appendix B.

B A Conditional Quotient-Optimization Separation

B.1 Quotient Geometry

For one RLB layer, suppress the layer index and write

$$z_g = A_g x, \quad (20)$$
$$r_g = \sqrt{m^{-1}\|z_g\|_2^2 + \epsilon_{\mathrm{rms}}}, \quad (21)$$
$$u_g = z_g / r_g, \quad (22)$$
$$h_g = r_g R_g(u_g), \quad (23)$$
$$y = B \operatorname{concat}_{g=1}^G h_g. \quad (24)$$

For $a_g > 0$, define

$$T_a(A_g, B_g) = (a_g A_g, a_g^{-1} B_g), \qquad g = 1, \dots, G. \quad (25)$$

Equivalently, with $s_g = \log a_g$,

$$T_s(A_g, B_g) = (e^{s_g} A_g, e^{-s_g} B_g). \quad (26)$$

Let $\mathcal{P}$ be the local RLB-matrix parameter space, let $\mathcal{G} = (\mathbb{R}, +)^G$, and let $\mathcal{Q} = \mathcal{P}/\mathcal{G}$.

Lemma B.1 (Exact positive gauge). If $\epsilon_{\mathrm{rms}} = 0$ and each zero-radius group, if present, is evaluated by a bounded convention with $h_g = 0$, then $T_s(A, B)$ and (A, B) represent the same layer map for all $s \in \mathbb{R}^G$. In particular, the claim holds on any input for which all group radii are nonzero.

Proof. For a group with $r_g > 0$, $A'_g = e^{s_g} A_g$ gives $z'_g = e^{s_g} z_g$. Since $\epsilon_{\mathrm{rms}} = 0$, $r'_g = e^{s_g} r_g$, hence $u'_g = u_g$ and $h'_g = e^{s_g} h_g$. The corresponding output block satisfies $B'_g h'_g = e^{-s_g} B_g e^{s_g} h_g = B_g h_g$. If $r_g = 0$, then $z_g = 0$ and also $z'_g = 0$; by the zero-radius convention both group features are zero, so the output block is again unchanged. Summing over groups gives the claim. □

Proposition B.1 (Stabilized gauge error). Fix $|s_g| \le S$ and $\epsilon_0 > 0$, and let f_ϵ denote the layer map with radius $r_g = \sqrt{m^{-1} \|z_g\|_2^2 + \epsilon}$. Suppose that, on the considered local activation set,

$$0 < \sigma^2 \le m^{-1} \|z_g\|_2^2 \le M^2 < \infty \tag{27}$$

for every group, the rational maps are bounded and Lipschitz on the resulting normalized inputs uniformly for $0 \le \epsilon \le \epsilon_0$ and $|s_g| \le S$, and the output blocks B_g have bounded operator norm. Then there is a constant C independent of ϵ such that, for all $0 \le \epsilon \le \epsilon_0$,

$$\|f_\epsilon(T_s(A, B); x) - f_\epsilon(A, B; x)\|_2 \le C\epsilon. \tag{28}$$

Proof. It suffices to bound one group. Let $a = e^{s_g}$ and $S_z = m^{-1} \|z_g\|_2^2$. After applying T_s, the contribution of this group after the inverse scaling of B_g is governed by the radius $\sqrt{S_z + \epsilon/a^2}$, whereas the original radius is $\sqrt{S_z + \epsilon}$. Because a is bounded above and below and $S_z \ge \sigma^2$,

$$\left| \sqrt{S_z + \epsilon/a^2} - \sqrt{S_z + \epsilon} \right| \le C_1 \epsilon. \tag{29}$$

The normalized inputs differ by $O(\epsilon)$ by the same calculation applied to the reciprocal square-root factor. Boundedness and Lipschitzness of R_g, together with the bounded operator norm of B_g, therefore give an $O(\epsilon)$ bound for the group output. Summing over finitely many groups proves equation 28. □

Assumption B.1 (Local quotient chart). In a neighborhood of the trajectory, parameters can be written as

$$\theta = \theta(q, s), \qquad q \in \mathbb{R}^d, \qquad s \in \mathbb{R}^G, \tag{30}$$

where q is a local coordinate on $\mathcal{Q}$ and s is the log-gauge coordinate. The objective satisfies

$$L(\theta(q, s)) = F(q). \tag{31}$$

For exact gauge identities, $\epsilon_{\mathrm{rms}} = 0$. For nonzero RMS stabilization, Proposition B.1 gives a local $O(\epsilon_{\mathrm{rms}})$ gauge error under nondegenerate activations; the remaining implementation effects are included in the residual model of Assumption B.3.

Lemma B.2 (Gauge tangent orthogonality). Under Lemma B.1, for any differentiable loss depending on the RLB matrices only through the represented function, with Frobenius inner products,

$$\langle \nabla_{A_g} L, A_g \rangle - \langle \nabla_{B_g} L, B_g \rangle = 0. \tag{32}$$

Proof. Differentiate $L(e^{s_g} A_g, e^{-s_g} B_g) = L(A_g, B_g)$ at $s_g = 0$. □

B.2 Canonicalized Quotient Preconditioning

For group g, define

$$\gamma_g = \log \mathrm{rms}(A_g) - \log \mathrm{rms}(B_g). \tag{33}$$

The logarithms in this subsection are taken on true positive RMS quantities. Stabilized RMS floors preserve the lemma exactly only when inactive; otherwise the resulting discrepancy is part of the perturbation model in Assumption B.3. Given a target log-ratio τ_g, the unclipped rebalance is

$$\ell_g = \frac{1}{2}(\tau_g - \gamma_g), \qquad A_g^+ = e^{\ell_g} A_g, \qquad B_g^+ = e^{-\ell_g} B_g. \tag{34}$$

Lemma B.3 (RMS-ratio canonicalization). For true positive RMS values, the update in equation 34 satisfies

$$\log \mathrm{rms}(A_g^+) - \log \mathrm{rms}(B_g^+) = \tau_g. \tag{35}$$

Proof. By homogeneity of RMS under positive scaling,

$$\begin{aligned}\log \mathrm{rms}(A_g^+) - \log \mathrm{rms}(B_g^+) &= \log \mathrm{rms}(e^{\ell_g} A_g) - \log \mathrm{rms}(e^{-\ell_g} B_g) & (36)\\ &= \gamma_g + 2\ell_g = \tau_g. & (37)\end{aligned}$$

□

Lemma B.4 (Gauge-dependent raw quotient step). Let $w > 0$, $a = e^s\sqrt{w}$, $b = e^{-s}\sqrt{w}$, and $\varphi(w) = \frac{1}{2}(w-y)^2$. One Euclidean gradient-descent step in (a, b) with step size η gives

$$w^+ = w - 2\eta w \cosh(2s)(w - y) + O(\eta^2). \tag{38}$$

Proof. Since $\partial\varphi/\partial a = (w-y)b$ and $\partial\varphi/\partial b = (w-y)a$,

$$\begin{aligned}w^+ &= (a - \eta(w-y)b)(b - \eta(w-y)a) & (39)\\ &= w - \eta(w-y)(a^2+b^2) + O(\eta^2). & (40)\end{aligned}$$

Using $a^2 + b^2 = w(e^{2s} + e^{-2s}) = 2w\cosh(2s)$ gives equation 38. □

Assumption B.2 (Frozen quotient preconditioners for the quadratic specialization). During the local phase, MatrixPolicy and AdamW induce fixed positive definite quotient preconditioners P_{MP} and P_{AdamW} satisfying

$$\begin{aligned}q_{t+1}^{\mathrm{MP}} &= q_t^{\mathrm{MP}} - \eta_{\mathrm{MP}} P_{\mathrm{MP}} \nabla F(q_t^{\mathrm{MP}}), & (41)\\ q_{t+1}^{\mathrm{AdamW}} &= q_t^{\mathrm{AdamW}} - \eta_{\mathrm{AdamW}} P_{\mathrm{AdamW}} \nabla F(q_t^{\mathrm{AdamW}}). & (42)\end{aligned}$$

Here P_{MP} is the quotient preconditioner after gauge rebalance and P_{AdamW} is the quotient preconditioner induced by AdamW's raw-coordinate adaptive state.

B.3 Contraction and Perturbation Bounds

For $P \succ 0$, define the transformed quotient objective

$$\widetilde{F}_P(x) = F(q_\star + P^{1/2}x). \tag{43}$$

A preconditioned quotient step $q^+ = q - \eta P \nabla F(q)$ is equivalent to ordinary gradient descent on $\widetilde{F}_P$ in the coordinate $x = P^{-1/2}(q - q_\star)$.

Lemma B.5 (Pathwise contraction certificate). Let an ideal quotient trajectory for optimizer o have nonnegative gaps $Y_t^o = F(q_t^o) - F_\star \geq 0$ and satisfy

$$Y_{t+1}^o \leq \alpha_{o,t} Y_t^o, \qquad \alpha_{o,t} \geq 0, \tag{44}$$

for $0 \leq t < T$. Then

$$F(q_T^o) - F_\star \leq \left(\prod_{t=0}^{T-1} \alpha_{o,t}\right)\left(F(q_0^o) - F_\star\right). \tag{45}$$

The factors need not be below one at every step; a finite-horizon contraction follows whenever their product is below one.

Proof. Iterate equation 44. □

Proposition B.2 (Expected contraction certificate). Let q_t^o be a stochastic quotient trajectory adapted to a filtration $\mathcal{H}_t$, and set $Y_t^o = F(q_t^o) - F_\star \geq 0$. Suppose there are nonnegative $\mathcal{H}_t$-measurable factors $\alpha_{o,t}$ and deterministic upper bounds $\bar{\alpha}_{o,t}$ such that $\alpha_{o,t} \leq \bar{\alpha}_{o,t}$ almost surely and

$$\mathbb{E}\left[Y_{t+1}^o \mid \mathcal{H}_t\right] \leq \alpha_{o,t} Y_t^o \tag{46}$$

for $0 \leq t < T$. Then

$$\mathbb{E}\left[F(q_T^o) - F_\star\right] \leq \left(\prod_{t=0}^{T-1} \bar{\alpha}_{o,t}\right) \mathbb{E}\left[F(q_0^o) - F_\star\right]. \tag{47}$$

The factors may be state dependent, the initial point may be random, and no individual envelope has to be below one; finite-horizon expected contraction is controlled by the product of the deterministic envelopes. For two optimizers initialized from the same distribution, the smaller product gives the smaller certified expected finite-horizon bound.

Proof. Since Y_t^o and $\alpha_{o,t}$ are $\mathcal{H}_t$-measurable and nonnegative,

$$\mathbb{E}Y_{t+1}^o \leq \mathbb{E}\left[\alpha_{o,t} Y_t^o\right] \leq \bar{\alpha}_{o,t}\mathbb{E}Y_t^o. \tag{48}$$

Iterating proves equation 47. □

Lemma B.6 (One-step transformed PL decrease). Let $P \succ 0$, define $\widetilde{F}_P$ by equation 43, and set $x = P^{-1/2}(q - q_\star)$ and $x^+ = x - L_P^{-1}\nabla\widetilde{F}_P(x)$. Suppose the pathwise descent inequality

$$\widetilde{F}_P(x^+) \leq \widetilde{F}_P(x) - \frac{1}{2L_P}\left\|\nabla\widetilde{F}_P(x)\right\|_2^2 \tag{49}$$

and the pointwise PL inequality

$$\frac{1}{2}\left\|\nabla\widetilde{F}_P(x)\right\|_2^2 \geq \mu_P\big(\widetilde{F}_P(x) - F_\star\big), \qquad 0 < \mu_P \leq L_P, \tag{50}$$

hold at the current iterate. If

$$q^+ = q - L_P^{-1}P\nabla F(q), \tag{51}$$

then

$$F(q^+) - F_\star \leq \left(1 - \frac{\mu_P}{L_P}\right)\big(F(q) - F_\star\big). \tag{52}$$

The descent condition equation 49 is implied by L_P-smoothness of $\widetilde{F}_P$ on the segment from x to x^+, but global smoothness or convexity is not required.

Proof. The update in q is the gradient step $x^+ = x - L_P^{-1}\nabla\widetilde{F}_P(x)$. Combining equation 49 and equation 50 gives

$$\widetilde{F}_P(x^+) - F_\star \leq \left(1 - \frac{\mu_P}{L_P}\right)\big(\widetilde{F}_P(x) - F_\star\big). \tag{53}$$

Since $\widetilde{F}_P(x) = F(q)$ and $\widetilde{F}_P(x^+) = F(q^+)$, this proves equation 52. □

Proposition B.3 (Time-varying PL quotient bound). For each optimizer $o \in \{\mathrm{MP}, \mathrm{AdamW}\}$, suppose both ideal quotient trajectories start from the same quotient point q_0 and have the form

$$q_{t+1}^o = q_t^o - L_{o,t}^{-1}P_{o,t}\nabla F(q_t^o), \qquad P_{o,t} \succ 0, \tag{54}$$

and that Lemma B.6 applies at every step with constants $(L_{o,t}, \mu_{o,t})$. Define

$$\kappa_{o,t} = \frac{L_{o,t}}{\mu_{o,t}}, \qquad \Xi_o(T) = \prod_{t=0}^{T-1}\left(1 - \kappa_{o,t}^{-1}\right). \tag{55}$$

Then

$$F(q_T^o) - F_\star \leq \Xi_o(T)\big(F(q_0) - F_\star\big). \tag{56}$$

Consequently, if $\Xi_{\mathrm{MP}}(T) < \Xi_{\mathrm{AdamW}}(T)$, MatrixPolicy has the smaller certified finite-horizon PL contraction bound.

Proof. Lemma B.6 gives the pathwise certificate equation 44 with $\alpha_{o,t} = 1 - \kappa_{o,t}^{-1}$. Apply Lemma B.5. □

Corollary B.1 (Frozen PL quotient bound). Suppose the frozen quotient dynamics in Assumption B.2 hold. For $o \in \{\mathrm{MP}, \mathrm{AdamW}\}$, suppose Lemma B.6 applies along the transformed ideal trajectory with $P = P_o$ and fixed constants (L_o, μ_o). Let

$$\kappa_o = \frac{L_o}{\mu_o}, \qquad \chi_o = 1 - \kappa_o^{-1}. \tag{57}$$

If $\kappa_{\mathrm{MP}} < \kappa_{\mathrm{AdamW}}$ and $\eta_o = 1/L_o$, then

$$\chi_{\mathrm{MP}} < \chi_{\mathrm{AdamW}} < 1 \tag{58}$$

and, for every initial point whose two ideal trajectories remain in their local regions,

$$F(q_T^{\mathrm{MP}}) - F_\star \le \chi_{\mathrm{MP}}^T \big(F(q_0) - F_\star\big), \tag{59}$$

$$F(q_T^{\mathrm{AdamW}}) - F_\star \le \chi_{\mathrm{AdamW}}^T \big(F(q_0) - F_\star\big). \tag{60}$$

Proof. This is Proposition B.3 with constant $P_{o,t} = P_o$, $L_{o,t} = L_o$, and $\mu_{o,t} = \mu_o$. The map $\kappa \mapsto 1 - \kappa^{-1}$ is strictly increasing for $\kappa \ge 1$. □

Proposition B.4 (Time-varying quadratic spectral bound). Let

$$F(q) = F_\star + \frac{1}{2}(q - q_\star)^\top H (q - q_\star), \qquad H \succ 0, \tag{61}$$

and consider the time-varying preconditioned update

$$q_{t+1} = q_t - \eta_t P_t \nabla F(q_t). \tag{62}$$

Define

$$A_t = H^{1/2}(I - \eta_t P_t H) H^{-1/2}, \qquad \alpha_t = \|A_t\|_2^2 . \tag{63}$$

Then

$$F(q_T) - F_\star \le \left(\prod_{t=0}^{T-1} \alpha_t \right) \big(F(q_0) - F_\star\big). \tag{64}$$

Proof. Let $e_t = q_t - q_\star$ and $y_t = H^{1/2} e_t$. Since $\nabla F(q_t) = H e_t$,

$$y_{t+1} = H^{1/2}(I - \eta_t P_t H) H^{-1/2} y_t = A_t y_t. \tag{65}$$

Also $2(F(q_t) - F_\star) = \|y_t\|_2^2$. Hence

$$2(F(q_{t+1}) - F_\star) = \|A_t y_t\|_2^2 \le \alpha_t \, \|y_t\|_2^2 = 2\alpha_t (F(q_t) - F_\star). \tag{66}$$

Iterating this one-step inequality gives equation 64. □

Lemma B.7 (Preconditioned quadratic contraction). Let

$$F(q) = F_\star + \frac{1}{2}(q - q_\star)^\top H (q - q_\star), \qquad H \succ 0. \tag{67}$$

For $P \succ 0$, define

$$K(P) = P^{1/2} H P^{1/2}, \qquad \kappa(P; H) = \frac{\lambda_{\max}(K(P))}{\lambda_{\min}(K(P))}, \qquad \rho(P; H) = \frac{\kappa(P; H) - 1}{\kappa(P; H) + 1}. \tag{68}$$

With

$$\eta^\star(P) = \frac{2}{\lambda_{\max}(K(P)) + \lambda_{\min}(K(P))}, \tag{69}$$

the update $q_{t+1} = q_t - \eta^\star(P) P \nabla F(q_t)$ satisfies

$$F(q_T) - F_\star \le \rho(P; H)^{2T} \big(F(q_0) - F_\star\big), \tag{70}$$

and the bound is sharp.

Proof. Let $e_t = q_t - q_\star$ and $\widetilde{e}_t = P^{-1/2} e_t$. Then

$$\widetilde{e}_{t+1} = (I - \eta^\star(P) K(P))\,\widetilde{e}_t, \qquad 2(F(q_t) - F_\star) = \widetilde{e}_t^T K(P) \widetilde{e}_t. \tag{71}$$

Diagonalize $K(P) = U\Lambda U^T$ and set $v_t = U^T \widetilde{e}_t$. The largest multiplier satisfies

$$\max_i |1 - \eta^\star(P)\lambda_i| = \frac{\lambda_{\max}(K(P)) - \lambda_{\min}(K(P))}{\lambda_{\max}(K(P)) + \lambda_{\min}(K(P))} = \rho(P; H). \tag{72}$$

Thus

$$2(F(q_{t+1}) - F_\star) = \sum_i \lambda_i v_{t+1,i}^2 \tag{73}$$

$$\leq \rho(P; H)^2 \sum_i \lambda_i v_{t,i}^2 \tag{74}$$

$$= 2\rho(P; H)^2 (F(q_t) - F_\star). \tag{75}$$

Iteration gives equation 70. Equality holds when the transformed initial error $\widetilde{e}_0 = P^{-1/2}(q_0 - q_\star)$ is an eigenvector of $K(P)$ associated with $\lambda_{\min}(K(P))$ or $\lambda_{\max}(K(P))$. □

Theorem B.1 (Conditional worst-case optimization separation). *Suppose the frozen quotient dynamics in Assumption B.2 hold, and suppose F is the quadratic objective in Lemma B.7. Assume also that*

$$\kappa(P_{\mathrm{MP}}; H) < \kappa(P_{\mathrm{AdamW}}; H), \tag{76}$$

and set $\eta_{\mathrm{MP}} = \eta^\star(P_{\mathrm{MP}})$ and $\eta_{\mathrm{AdamW}} = \eta^\star(P_{\mathrm{AdamW}})$. For every integer $T \geq 1$, let U_T be a sufficiently small quotient neighborhood of $q_\star$ on which both ideal trajectories remain in the local chart through time T. Then

$$\sup_{\substack{q_0 \in U_T \\ q_0 \neq q_\star}} \frac{F(q_T^{\mathrm{MP}}) - F_\star}{F(q_0) - F_\star} = \rho(P_{\mathrm{MP}}; H)^{2T} < \rho(P_{\mathrm{AdamW}}; H)^{2T} = \sup_{\substack{q_0 \in U_T \\ q_0 \neq q_\star}} \frac{F(q_T^{\mathrm{AdamW}}) - F_\star}{F(q_0) - F_\star}. \tag{77}$$

Proof. The map $\kappa \mapsto (\kappa - 1)/(\kappa + 1)$ is strictly increasing for $\kappa > 0$. The strict condition-number inequality therefore gives $\rho(P_{\mathrm{MP}}; H) < \rho(P_{\mathrm{AdamW}}; H)$. Lemma B.7 gives the sharp worst-case factors. Since the quadratic ratios are invariant under scaling $q_0 - q_\star$, the extremal transformed-eigenvector initializations can be chosen inside any sufficiently small neighborhood U_T. □

Assumption B.3 (Uniform local perturbation bound). *For a fixed horizon T, let the ideal quotient recursions be*

$$q_{t+1}^o = \Psi_{o,t}(q_t^o), \qquad o \in \{\mathrm{MP}, \mathrm{AdamW}\}. \tag{78}$$

The implemented quotient iterates satisfy

$$q_{t+1}^{o,\mathrm{impl}} = \Psi_{o,t}(q_t^{o,\mathrm{impl}}) + r_t^o, \qquad \left\|r_t^{\mathrm{MP}}\right\|_2 + \left\|r_t^{\mathrm{AdamW}}\right\|_2 \leq \delta \tag{79}$$

for all $0 \leq t < T$. The residuals include the RMS floor, denominator floors, clipping, moment lag, weight decay, and nonquadratic Taylor remainders.

Proposition B.5 (Stability of a strict finite-time gap). *Fix $T \geq 1$ and suppose the idealized dynamics satisfy*

$$m_T := F(q_T^{\mathrm{AdamW}}) - F(q_T^{\mathrm{MP}}) > 0. \tag{80}$$

Assume the implemented and ideal recursions use the same initial points, and assume the ideal trajectories lie in a compact set $K \subset \mathrm{int}(U)$, where U is a compact neighborhood on which F and all $\Psi_{o,t}$ are Lipschitz. Then there exists $\delta_0 > 0$ such that, whenever Assumption B.3 holds with $\delta < \delta_0$, the implemented trajectories remain in U through time T and

$$F(q_T^{\mathrm{AdamW,impl}}) - F(q_T^{\mathrm{MP,impl}}) > \frac{m_T}{2} > 0. \tag{81}$$

Proof. Let C_Ψ be a common Lipschitz constant on U for the finite family of ideal one-step maps $\{\Psi_{o,t} : o \in \{\mathrm{MP}, \mathrm{AdamW}\}, 0 \leq t < T\}$. Let $r_{\mathrm{stay}} = \mathrm{dist}(K, U^c) > 0$. By induction, as long as the implemented trajectories remain in U,

$$\left\| q_t^{\mathrm{MP,impl}} - q_t^{\mathrm{MP}} \right\| + \left\| q_t^{\mathrm{AdamW,impl}} - q_t^{\mathrm{AdamW}} \right\| \leq C_T \delta \tag{82}$$

for a constant C_T depending only on T and C_Ψ. Taking $\delta \leq r_{\mathrm{stay}}/(2C_T)$ keeps the implemented trajectories in U and closes the induction. If C_F is a Lipschitz constant for F on U, then

$$\left| F(q_T^{\mathrm{MP,impl}}) - F(q_T^{\mathrm{MP}}) \right| + \left| F(q_T^{\mathrm{AdamW,impl}}) - F(q_T^{\mathrm{AdamW}}) \right| \leq C_F C_T \delta. \tag{83}$$

Taking

$$\delta_0 = \min \left\{ \frac{m_T}{2C_F C_T}, \frac{r_{\mathrm{stay}}}{2C_T} \right\} \tag{84}$$

gives the claim. □